

①

$$w = 0.5$$
$$b = -1$$

$$x = 2$$

$$z = wx + b$$

$$z = 0.5 \times 2 - 1 = \underline{\underline{0}}$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-0}} = \underline{\underline{\frac{1}{2}}}$$

②

$$\hat{y} = 0.8$$

$$y = 1$$

$$L = -(1) \ln(0.8) = \underline{\underline{-\ln(0.8)}}$$

$$\textcircled{3} \quad L = -y \ln(\hat{y}) - (1-y) \ln(1-\hat{y})$$

$$\hat{y} = \sigma(z)$$

$$\frac{dL}{dz} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dz}$$



$$\frac{dL}{d\hat{y}} = \frac{-y}{\hat{y}} + \frac{1-y}{1-\hat{y}} = \frac{-y + \cancel{\hat{y}} + \hat{y} \cancel{1}}{\hat{y} \cdot (1-\hat{y})}$$

$$= \frac{\hat{y} - y}{\hat{y}(1-\hat{y})}$$

$$\frac{d\hat{y}}{dz} = \sigma(z)(1-\sigma(z)) = \hat{y}(1-\hat{y})$$

$$\frac{dL}{dz} = \frac{\hat{y} - y}{\cancel{\hat{y}(1-\hat{y})}} \cdot \cancel{\hat{y}(1-\hat{y})} = \underline{\underline{\hat{y} - y}}$$

$$\textcircled{1} \frac{\partial L}{\partial z} = \hat{y} - y$$

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial z} \cdot \frac{\partial z}{\partial w}$$

$$\frac{\partial z}{\partial w} = x$$

$$\bullet \frac{\partial L}{\partial w} = (\hat{y} - y) x$$

$$\bullet \frac{\partial L}{\partial b} = (\hat{y} - y) \cdot 1$$

$$\Rightarrow \hat{y} = 0.7 \quad y = 1 \quad x = 3$$

$$\text{i)} (-0.3) \cdot 3 = -0.9$$

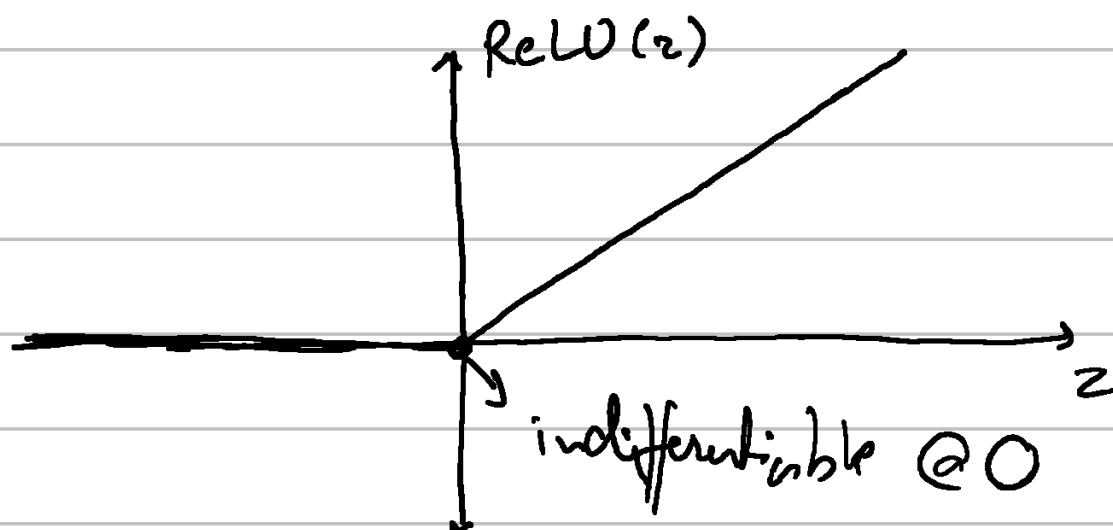
$$\text{ii)} (-0.3) \cdot 1 = \underline{\underline{-0.3}}$$

$$\textcircled{5} \quad w_i \leftarrow w - \alpha \frac{\partial L}{\partial w}$$

$$\begin{aligned} w_i &= 0.5 - (0.1) (-0.9) \\ &= 0.5 + 0.09 \\ &= \underline{\underline{0.59}} \end{aligned}$$

$$\begin{aligned} b_i &= -1 - (0.1) \cdot (-0.9) \\ &= -1 + 0.09 = \underline{\underline{-0.91}} \end{aligned}$$

check also $\rightarrow$ vanishing gradient problem.
 $\textcircled{6}$ $\text{ReLU}(z) = \max(0, z)$



$$\text{ReLU}'(z) = \begin{cases} 1 & z > 0 \\ 0 & z < 0 \\ \text{undef} & z = 0 \end{cases}$$

In really deep networks, the differential makes calculations quite fast as it completely either eliminates or keeps a term.

Whereas in sigmoids we might do a calculation for something that has very little contribution in the output!

⇒ So, we can rely on the number of neurons instead of the floating point precision of sigmoid.

$$\textcircled{7} \quad \frac{d}{dz} \tanh(z) = 1 - \tanh^2(z)$$

$$\Rightarrow \underline{\tanh(z) = 2\sigma(2z) - 1}$$

$$1 - \tanh^2(z) = 2 \cdot 2 \cdot (\sigma(2z) \cdot (1 - \sigma(2z)))$$

$$1 - (2\sigma(2z) - 1)^2 = 4 \cdot \sigma(2z)(1 - \sigma(2z))$$

$$(1 + 2\sigma(2z) - 1)(1 - 2\sigma(2z) + 1)$$

$$\underline{LHS = RHS}$$

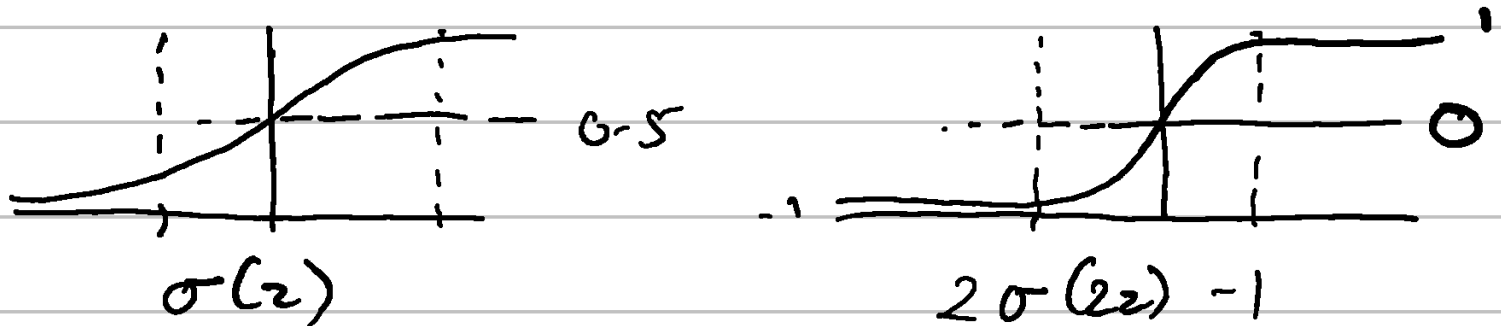
output range of .

$$\sigma(z) \rightarrow (0, 1)$$

$$\tanh(z) \rightarrow (-1, 1)$$

X From the looks of it, $\tanh(z)$ seems to be reaching asymptotic bounds faster than sigmoid. X

kinda like



Actual reason

The gradients passed to hidden layers have a balanced sign distribution which can make learning faster.

In practice, $\tanh$ is usually preferred for hidden layers & σ is reserved for binary classifier output layers.

$$\textcircled{8} \quad \hat{y} = \omega x + b$$

$$L = (\hat{y} - y)^2$$

$$\frac{\partial L}{\partial \omega} = 2 \cdot (\hat{y} - y) \cdot x$$

$$\frac{\partial L}{\partial b} = 2 \cdot (\hat{y} - y)$$

$$\omega = 1$$

$$b = 0$$

$$x = 3$$

$$y = 2$$

$$\hat{y} = 1 \cdot 3 + 0 = 3$$

$$\frac{\partial L}{\partial \omega} = 2 \cdot (3 - 2) \cdot 3 = \underline{\underline{6}}$$

$$\underline{\underline{\frac{\partial L}{\partial b} = 2}}$$

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$$z_1 = w_1 x + b_1$$

$$a_1 = \sigma(z_1)$$

$$z_2 = w_2 a_1 + b_2$$

$$\hat{y} = \sigma(z_2)$$

$$y = 1$$

$$\begin{aligned} \frac{\partial L}{\partial w_1} &= \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_2} \cdot \frac{\partial z_2}{\partial a_1} \cdot \frac{\partial a_1}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_1} \\ &= (\hat{y} - y) \cdot w_2 \cdot a_1 \cdot (1 - a_1) \cdot x \end{aligned}$$

10 full forward

$$z_1 = 2 \cdot 1 + 0 = 2$$

$$a_1 = \sigma(z_1) = \sigma(2) = 0.881$$

$$z_2 = 1 \cdot (0.881) + 0 = 0.881$$

$$\hat{y} = \sigma(0.881) = 0.707$$

$$h = - (1) \ln(0.707) = -\ln(0.707)$$

Backprop

$$\begin{aligned} \frac{\partial L}{\partial w_1} &= (\hat{y} - y) \cdot w_2 \cdot a_1 (1 - a_1) \cdot x \\ &= (0.707 - 1) \cdot 1 \cdot (0.881)(1 - 0.881) \cdot 1 \\ &= (-0.293) \cdot 0.881 \cdot (0.119) \\ &= \underline{\underline{-0.03071}} \end{aligned}$$

$$\begin{aligned} \frac{\partial L}{\partial w_2} &= (\hat{y} - y) \cdot a_1 = -0.293 \cdot 0.881 \\ &= \underline{\underline{-0.2581}} \end{aligned}$$

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$$a_1 = \sigma(\omega_1 x)$$

$$a_2 = \sigma(\omega_2 a_1)$$

$$a_3 = \sigma(\omega_3 a_2) = \hat{y}$$

$$L = (a_3 - y)^2$$

$$\frac{\partial L}{\partial \omega_1} = \frac{\partial L}{\partial a_3} \cdot \frac{\partial a_3}{\partial a_2} \cdot \frac{\partial a_2}{\partial a_1} \cdot \frac{\partial a_1}{\partial \omega_1}$$

$$= 2 \cdot (a_3 - y) \cdot \overset{\omega_3}{a_3(1-a_3)} \cdot \overset{\omega_2}{a_2(1-a_2)} \cdot a_1(1-a_1) \cdot x$$

$$= 2 \cdot (a_3 - y) \cdot a_3 \omega_3 (1-a_3) \cdot a_2 \omega_2 (1-a_2) \cdot a_1 x (1-a_1)$$

$$\textcircled{12} \quad \sigma'(z) = \sigma(z)(1 - \sigma(z)) \leq \frac{1}{4}$$

$$\frac{\partial L}{\partial w_1} = 2 \cdot (a_3 - y) \cdot w_3 \cdot \sigma'(z_3) \cdot w_2 \cdot \sigma'(z_2) \cdot x \cdot \sigma'(z_1)$$

$$\frac{\partial L}{\partial w_1} \leq 2 \cdot (a_3 - y) \cdot w_3 \cdot w_2 \cdot x \cdot \left(\frac{1}{4}\right)^3$$

Now, as the layers grow,
 $\left(\frac{1}{4}\right)^k$ will keep on diminishing
 and "the gradient will lose its
 significance".

This is called the vanishing
 gradient problem. This is solved by
 using ReLU activation function.

(18)

$$z_1 = w_1 x + b_1 = 2$$

$$a_1 = \text{ReLU}(z_1) = 2$$

$$z_2 = w_2 a_1 + b_2 = 2$$

$$\hat{y} = \sigma(z_2) = 0.881$$

$$y = 1$$

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_2} \cdot \frac{\partial z_2}{\partial a_1} \cdot \frac{\partial a_1}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_1}$$

$$\Rightarrow (\hat{y} - y) \cdot w_2 \cdot \text{ReLU}'(z_1) \cdot x$$

Putting in the vals.

$$\Rightarrow (0.881 - 1) \cdot (1) \cdot (1) \cdot (1)$$

$$= -0.119$$

larger magnitude than σ for gradient.